

Basic Principles of Medicine 2
Module: Digestion and Metabolism

Absorption in the
Gastro-Intestinal Tract

Read before lecture 39

Department of Physiology & Biochemistry

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Sodium uptake & secretion

- Universal endothelial absorption/secretion driven by Na/K-ATPase creates strong inward Na^+ gradient (figure panel B)
- Basolateral primary active secretion of Na^+ into ECF/blood via Na/K-ATPase
- Resulting low intracellular Na^+ allows for apical facilitated diffusion of Na^+ (i.e. uptake) into the cells via Na^+ channels
- ENaC does not play a large role in enteric apical Na^+ uptake
- Instead, Na^+ gradient is used for secondary co-transport of other molecules:
 - Monosaccharides: SGLT-1
 - NHE (Na/H exchanger): H^+
 - Amino acid transporters

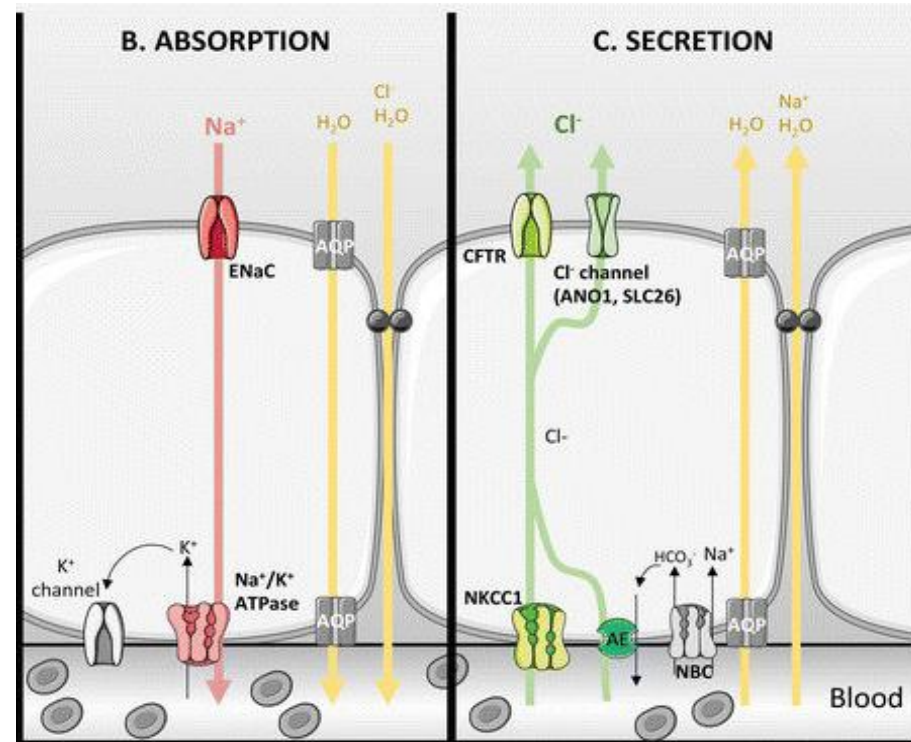
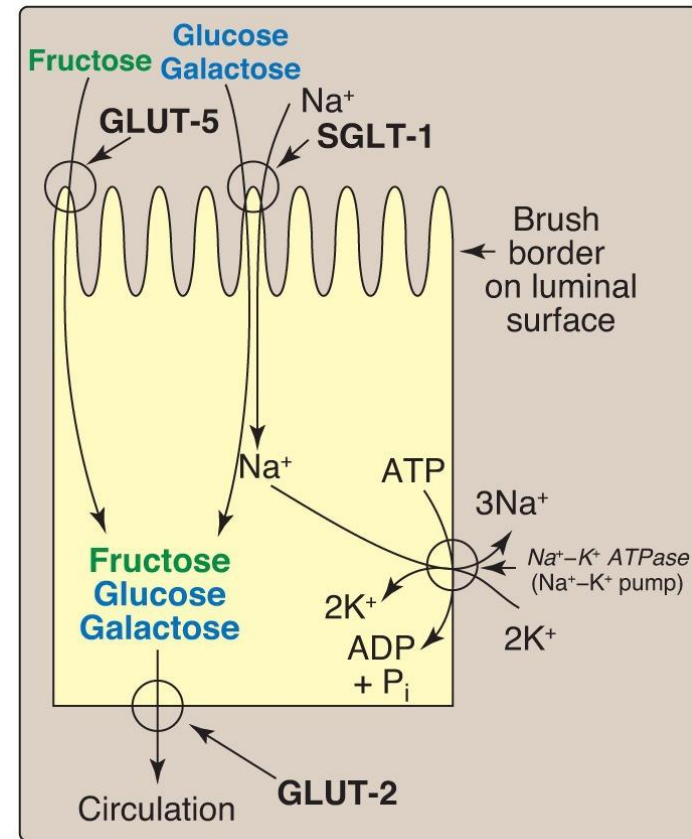


Image from Saint-Criq, V., Gray, M.A. Role of CFTR in epithelial physiology. *Cell. Mol. Life Sci.* 74, 93–115 (2017).
<https://doi.org/10.1007/s00018-016-2391-y>

Carbohydrates

- only Monosaccharides are absorbed
- fructose enters the intestinal mucosal cells via facilitated transport by Glut-5
- glucose and galactose enter the intestinal mucosal cells via SGLT-1, using secondary active co-transport with sodium ions
- all three monosaccharides are released by facilitated transport by GLUT-2 into the portal vein

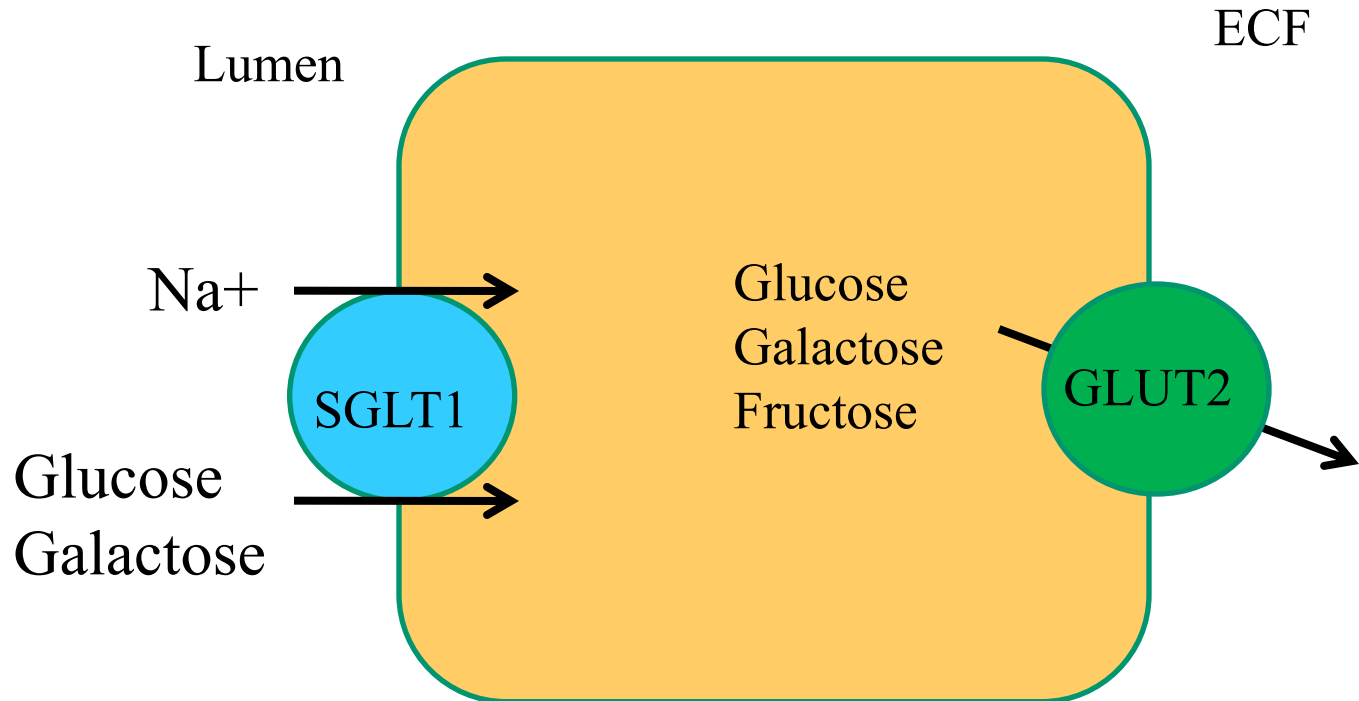


Concentration-dependent apical GLUT2 recruitment

- Apical transporters
 - GLUT5
 - fructose facilitated diffusion
 - SGLT1
 - uses Na^+ gradient (secondary active) to absorb glucose and galactose from lumen
 - 260 H_2O in per cycle $\rightarrow$ water absorption
 - 2 Na^+ in per glucose $\rightarrow$ depolarization
 - GLUT2
 - not normally present in apical membrane
 - induced to move into apical membrane by SGLT1-mediated depolarization at high luminal glucose levels: concentration-dependent recruitment
 - allows passive diffusion of glucose and galactose from lumen into epithelial cells, which is advantageous at high luminal concentrations when a strong concentration gradient drives diffusion in the desired direction
- Basolateral
 - GLUT2
 - facilitated diffusion into ECF/blood following the concentration gradient

Oral Rehydration Therapy: Solution with Na⁺ and Glucose

H₂O follows by osmosis



Absence of SGLT 1 transporter:

Glucose and Galactose malabsorption
→ Osmotic retention of water in GIT
→ Diarrhea

Water absorption

- Paracellular (passive)
 - H_2O follows NaCl osmotically (facilitated diffusion)
- Transcellular:
 - SGLT-1:
 - Secondary active (using Na^+ gradient)
 - SGLT-1 moves 260 water molecules into the cell per cycle
 - Calculated to account for up to ~5 L/day of water absorption!
 - Na^+ and glucose uptake adds to osmotic gradient for water absorption
 - Key pathway for oral rehydration solutions
 - facilitated diffusion via aquaporins
 - passive
 - driven by NaCl osmotic gradient like paracellular pathway

Secretion-absorption balance

- Key GIT fluid secretion sites:
 - Saliva (~1.5 L/day)
 - Stomach (~ 2 L/day)
 - Pancreas (~ 1.5 L/day)
 - Bile (~ 0.5 L/day)
 - Small intestine (~ 1 L/day)
- Key absorption sites:
 - Small intestine (~ 6.5 L/day)
 - Colon (~ 1.9 L/day)

GI fluid balance

SECRECTIONS

Diet: 1 – 2 L
Saliva: 1.5 L

Gastric Juice: 2.0 L

Pancreatic Juice: 1.5 L

Bile 0.5 L

Small Intestine 1 L

Colon

Fluid Load to SI =
7 – 9 L / day

ABSORPTION

6.5 L Small Intestine

2 L

1.9 L Colon

Stool water:
0.1 L

Fluid Absorbed =
98%

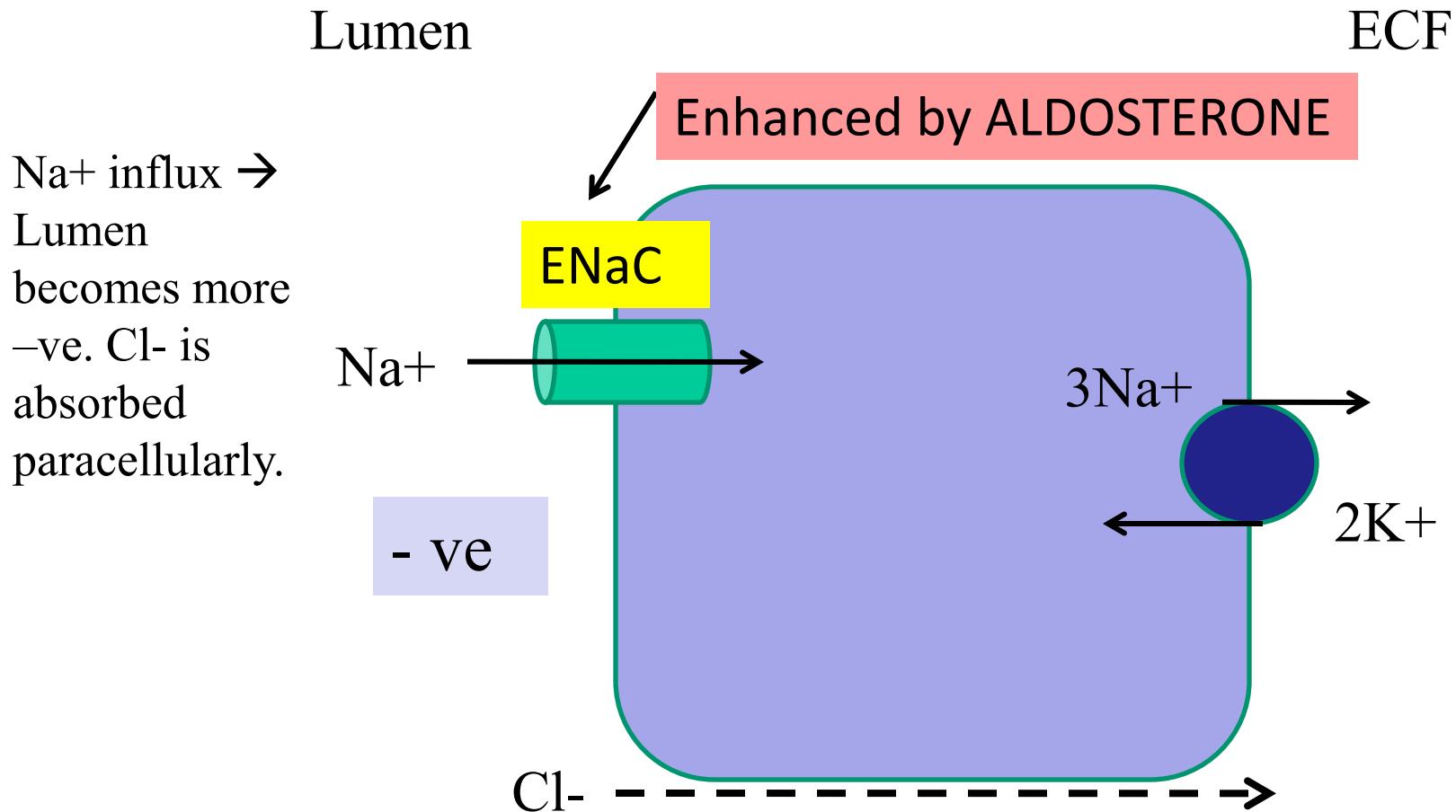
Intestinal solute and H₂O balance

- Secretory mechanisms are located in crypts
- Absorption mechanisms are located in the villi
- Cl⁻ is the primary ion secreted (see section on chloride below)
- Na⁺ secretion follows Cl⁻ by electrical attraction
- H₂O follows osmotically

Chloride uptake

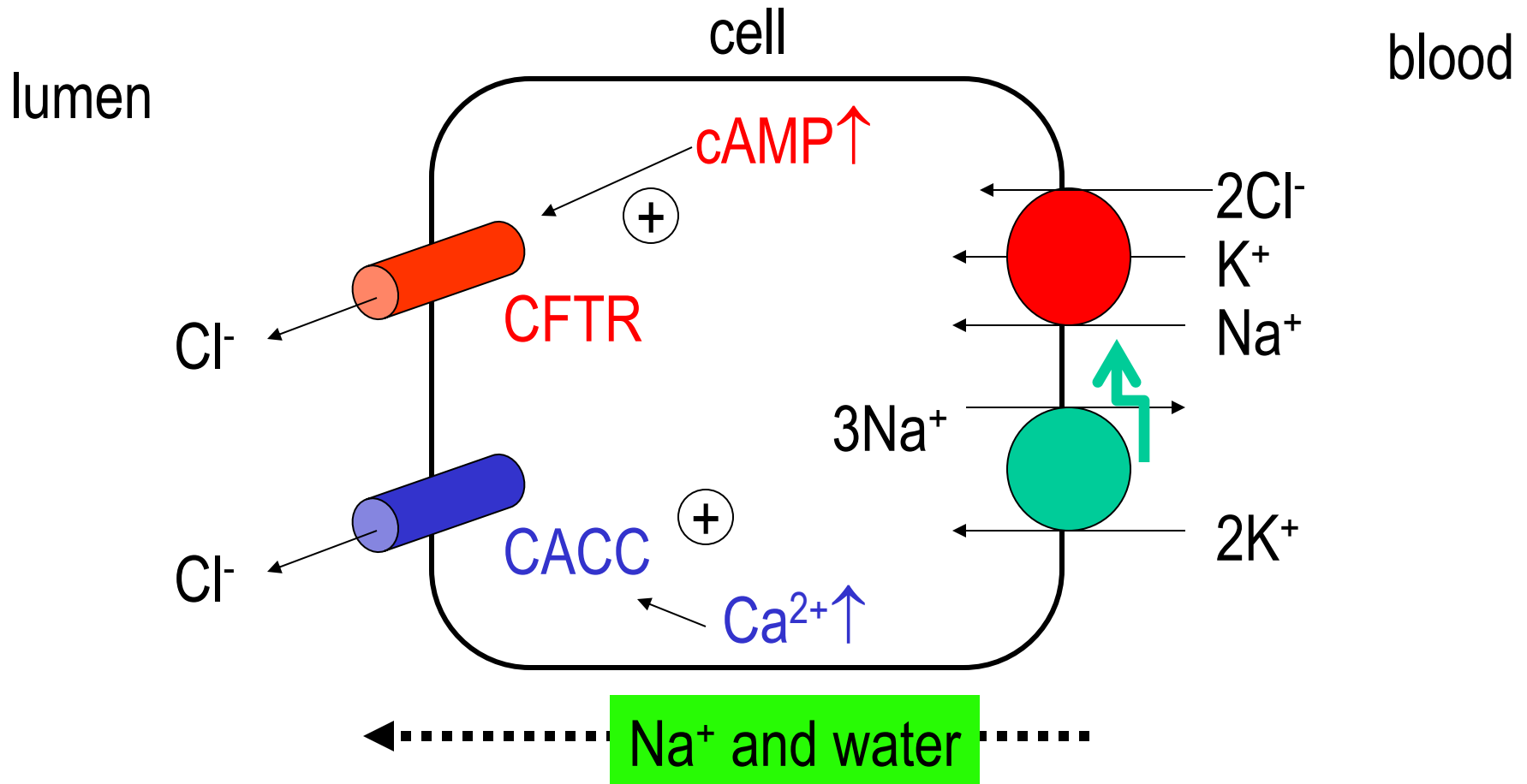
- 3 main uptake routes:
 - Paracellular
 - Passive
 - Driven by Na^+ absorption (see next slide)
 - Mainly in small intestine
 - Transcellular
 - Via $\text{Cl}^-/\text{HCO}_3^-$ exchanger
 - Linked to Na^+ absorption via NHE (Na^+/H^+ exchanger)
 - Net result is NaCl uptake and $\text{H}^+ + \text{HCO}_3^-$ secretion
 - Electrically and chemically neutral
 - Mainly in ileum and colon

Distal colon NaCl reabsorption



When body Na^+ is low, aldosterone stimulates the ENaC pump to promote Na^+ retention (similar in the kidney).

Cl⁻ Transport Drives Fluid Secretion From Crypts



NaCl Secretion in SI

Na⁺-K⁺ pump in basolateral membrane



Low intracellular [Na⁺]



Provides the energy for uptake of Cl⁻ into the cell via the NKCC transporter in basolateral membrane



[Cl⁻]_i increase so that Cl can exit passively through Cl channels in the apical membrane



Lumen becomes more negative



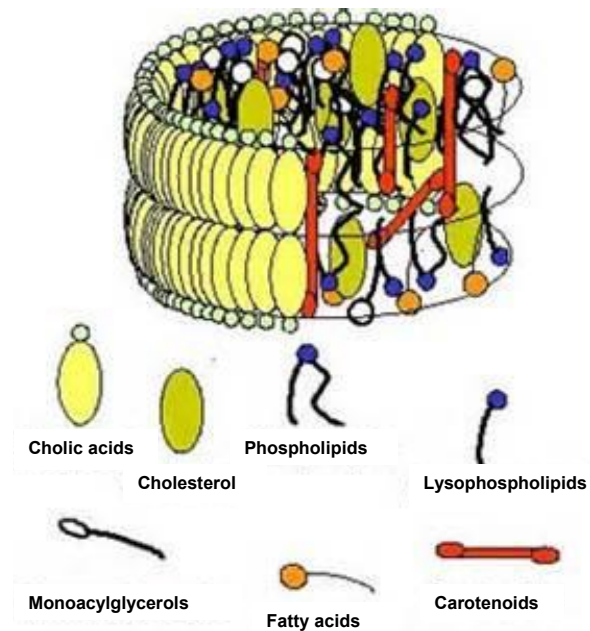
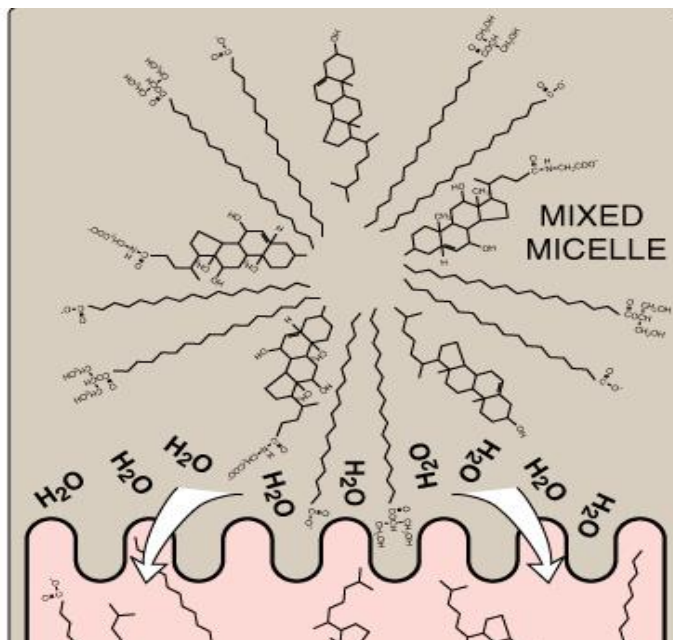
Attracts Na⁺ into the lumen down an electrical gradient



H₂O follows passively by osmosis

Lipids

Bile salts form mixed micelles for uptake of digested **lipids** into intestinal **mucosal cells**.



Amino Acids

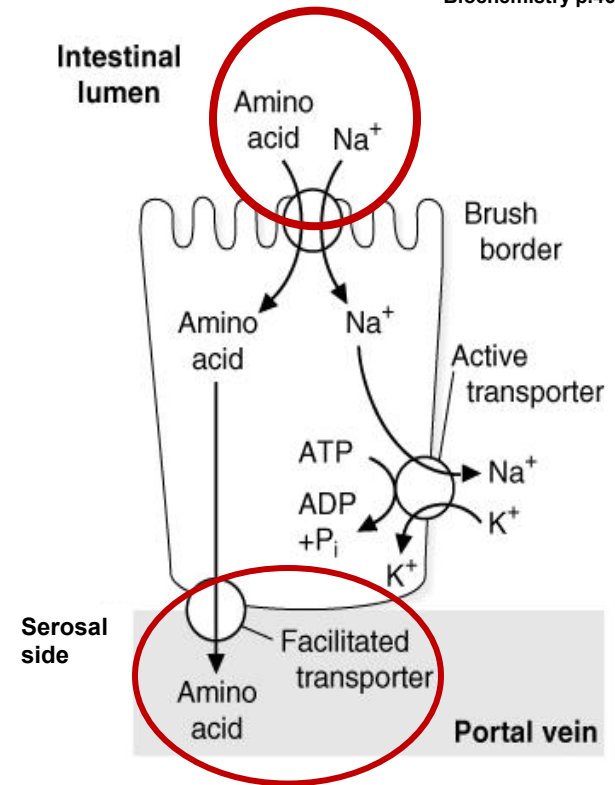
Absorption of dietary amino acids

Mark's Basic Medical
Biochemistry p.466

The **uptake** of dietary amino acids is mainly performed by **secondary active transport** with **cotransport** of sodium ions.

The **transporters** are **specific** for a **group** of **amino acids** which can be **overlapping**.

The **release** into the **portal vein** is by **facilitated transport**.



Further Reading

Review article with good overview and detail on micronutrients:

Kiela, P. R., & Ghishan, F. K. (2016). Physiology of Intestinal Absorption and Secretion. *Best Practice & Research Clinical Gastroenterology*, 30(2), 145–159. <https://doi.org/10.1016/j.bpg.2016.02.007>